

Multi-Agent AI Engineering Curriculum

A self-study curriculum for building production-grade multi-agent AI systems — 19 documents and 6 real projects, from Python fundamentals to a deployed, tested, multi-agent platform.

This is not a theory course. Every document pairs its concepts with hands-on practice exercises (each with progressive hints and full worked solutions), and every project is real, runnable code you build yourself, step by step.

How this curriculum is structured

- **19 documents** (01_python_foundations/ → 19_mlops_llmops/) — each one teaches a topic through simple-English "Core Concepts," then a set of practice exercises with a 2-hint, 3-depth (Basic/Intermediate/Advanced) hint system and full solutions, then a Build Task that feeds directly into the projects below.
- **6 projects** (project_1_beginner_llm_app/ → project_6_codeguard_bonus/) — real multi-agent systems you build across the documents, growing from a 2-agent terminal app to a tested, deployed, production-ready platform.
- Every document and project is delivered as both a README.md (source) and a matching README.pdf (for comfortable reading), with fully clickable internal links between exercises, hints, and solutions.

Documents

#	Document	Feeds into
01	Python & Engineering Foundations	—
02	APIs, HTTP, JSON & Backend Basics	—
03	LLM Basics	—
04	OpenAI API	Project 1
05	LangChain Basics	—
06	Tools & Function Calling	—
07	AI Agents	Project 2
08	RAG	—
—	Async Python (checkpoint)	—

#	Document	Feeds into
09	LangGraph	—
10	Agent Workflows	Project 3
11	Multi-Agent Systems	Project 4
12	Production AI Engineering	—
13	Testing, Evaluation & Observability	—
14	Debugging Lab	—
15	Five Projects (reference index)	—
16	System Design & Architecture	—
17	Interview Preparation	—
18	Final Production Multi-Agent Capstone	Project 5
19	MLOps & LLMOps	—

Projects

#	Project	Agents	Built during	Proves
1	SupportDesk: Concierge & Triage	2	Docs 01-04	You can talk to a model correctly and safely, and route between two simple agents
2	ResearchHand: Worker & Verifier	2	Docs 05-07	You can give a model real abilities, check its choices, and add an independent check
3	DocuMind: Retriever, Reasoner & Approval	3	Docs 08-10	You can build agents as a clear, checkable graph, with real handoffs
4	ContentForge: Supervisor-Led Team	5	Doc 11	You can split a problem across specialist agents, with a supervisor routing on the fly
5	ContentForge Pro: Production Platform	5 (production-ready)	Docs 12-19	You can ship, test, secure, watch, version, and defend the design of #4
6 (bonus)	CodeGuard: PR-Review Team	4	All of Docs 01-19, applied again	You can apply the whole architecture to a new problem, with less help

Agent count grows deliberately across the arc ($2 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 5$) — each project adds one new idea on top of the last, not several at once.

How to use this

1. Start at [01_python_foundations/](#) and work through the documents in order — each one lists its prerequisites at the top.
2. Read a document's Core Concepts, then do its Practice Exercises. Try each one yourself before opening its hints — use Hint 1, then Hint 2 only if you're still stuck, and the full solution only as a last resort.
3. When a document's Build Task is done, check [15_five_projects_index/](#) and start the project it unlocks.
4. Track your own progress in [PROGRESS.md](#).
5. See [CURRICULUM.md](#) for the full syllabus, rationale, and document-by-document breakdown.

Building the PDFs

Every `README.md` has a matching `README.pdf` with clickable internal links. If you edit any `.md` file, regenerate its PDF with the build tool in `tools/`:

```
cd tools
python3 build_pdfs.py                # rebuild everything
python3 build_pdfs.py ../01_python_foundations/README.md # rebuild just one file
python3 check_links.py               # verify every internal link resolves correctly
```

`watch_pdfs.py` can also be left running to rebuild automatically whenever a `.md` file changes.